ALY MAKLAD

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SUMMARY

Recent Computer Science graduate holding a dual Bachelor's degree from Ain Shams University and the University of East London, specializing in Artificial Intelligence. Hands-on experience across the full machine learning lifecycle - model development, data preprocessing, experimentation, and performance evaluation - using Python, PyTorch, and TensorFlow. Research background in fine-tuning LLMs (Phi-3, QLoRA) and building multimodal deep learning pipelines, with additional project-based training in LLM-powered AI agents (LangChain, RAG, tool-using agents). Experienced with SQL, Git, and full-stack development, and eager to apply a strong statistical and algorithmic foundation to real-world ML problems.

EXPERIENCE

Master AI Agents & LLMs - Hands-On Training Program, Digital Hub (D-Hub) & Orange Digital Center

Jun/20226 - Jul/2026 | Cairo, Egypt

LangChain | Flowise | LLMs | Prompt Engineering | RAG | Multi-Agent Systems | API Automation

- Completed a project-based training on AI agent architecture and the LLM ecosystem, covering prompt engineering, tool-using agents (web, math, APIs), memory and context management, and Retrieval-Augmented Generation over PDFs and Google Drive using LangChain and Flowise.
- Built and deployed a mid-term AI assistant that answers questions from PDFs and Google Sheets, and a final multi-agent system that retrieves data, summarizes it, and sends automated email/report outputs through a live web interface.

AI Research Intern, Manipal Institute of Technology *Jul/2025 - May/2026 | Manipal, India*

Machine Learning | Model Development | Medical Imaging | Vision-Language Models | LLMs

- Designed and trained a multimodal radiology report generation model on CT imaging data, preprocessing large-scale medical imaging datasets and training a MedicalNet-initialized ResNet50 encoder via MoCo self-supervised learning, aligning visual and text representations with CLIP-based contrastive learning.
- Built a custom image-to-text projection module and fine-tuned Phi-3 with 4-bit QLoRA under limited GPU resources (RTX 4070 8GB); evaluated model performance through iterative experiments, achieving a BERTScore-F1 of 0.568 and ROUGE-L of 0.146.

Full-Stack Developer, Notopia *Oct/2024 - Jan/2025 | Cairo, Egypt*

Python-adjacent Backend | Flutter | NestJS | REST APIs | Databases

- Contributed to a mobile e-commerce application for online stationery, building user-facing features in Flutter and backend services and REST APIs in NestJS.
- Implemented application features, integrated APIs, and managed database interactions (MongoDB) across the full stack.
- Collaborated with the team to deliver a scalable, production-ready product, gaining hands-on experience from feature design through deployment.

EDUCATION

Ain Shams University & University of East London, Dual Bachelor Degree in Computer Science *Sep/2022 - Jul/2026*

Cairo, Egypt

PROJECTS

Job Application Agent — Autonomous Job Search & Application System

LangGraph | LLMs (Ollama/Gemini/Groq) | Multi-Agent Orchestration | FastAPI | SQLAlchemy/SQLite

- Built an autonomous job-application agent that searches job boards for relevant openings, evaluates fit against the user's CV, tailors the CV when needed, and applies or drafts applications for review, freeing the user from manually searching and applying to jobs.
- Designed a multi-agent system of 8 specialized AI agents (search, ATS scoring, CV rewriting, apply-decision, email, reporting, news digest, skill-gap analysis) coordinated by a LangGraph orchestrator that autonomously routes each job through scoring, tailoring, and apply/draft decisions with no manual intervention.

- Automated the entire job-search lifecycle end-to-end via a scheduler running unattended daily/weekly: agents search multiple job boards, score fit using a hybrid keyword + LLM-judgment agent, auto-tailor CVs for low-fit roles, apply or draft applications through a whitelist-gated apply agent, and dispatch automated email and Telegram reports.

AI-Powered Course Material Chatbot (RAG)

NLP | LLMs | Retrieval-Augmented Generation | Vector Databases | FastAPI | Docker

- Built and shipped a complete chatbot app letting students upload course materials (PDFs, slides, notes, textbooks) and ask natural-language questions, with answers grounded in the uploaded content.
- Built the backend with FastAPI and containerized the app with Docker; combined vector database retrieval with an LLM to generate accurate, cited, document-grounded answers.

Shifaa - Hospital Monitoring Platform (Graduation Project)

Computer Vision | Deep Learning | Pose Estimation | Real-Time Systems

- Contributed to Shifaa, a privacy-first, on-premises hospital monitoring platform combining a grounded medical AI assistant with real-time cardiac arrhythmia, fall, and seizure detection behind a single role-aware interface.
- Built the real-time patient fall detection system using YOLO tracking, MobileNetV3 role classification, MediaPipe pose estimation, and CTR-GCN motion classification; evaluated performance at 91.59% accuracy, 91.22% precision, 98.83% recall, 94.87% F1, and 95.98% ROC AUC.

Object Detection & Tracking Optimization for Drone Systems (AIC Challenge)

Computer Vision | Deep Learning | Object Tracking | Model Optimization

- Developed a lightweight drone tracking model achieving 10ms latency (100 FPS), well under the 30ms constraint, within strict parameter limits (≤20 GFLOPs, ≤10M parameters).
- Applied pruning and TensorRT optimization to reach 85.6% Precision@20px and 0.698 Mean IoU across 31,700+ frames with zero sequence failures.

Food & Fruit Recognition with Calorie Estimation

Computer Vision | Deep Learning | Few-Shot Learning | Transformers

- Built a few-shot food recognition system using Prototypical Networks and Swin Transformer backbones, achieving 76.31% ± 0.56% accuracy in 5-way 10-shot evaluation with calorie estimation support.

EXTRACURRICULAR ACTIVITIES

- IGTa Member, AIESEC in Ain Shams University - Supported international talent exchange programs and recruitment coordination.
- Computer Vision Member, APPLAI in Ain Shams University - Conducted computer vision workshops and collaborated on AI-driven projects.
- HR Member, ACM ASCIS in Ain Shams University - Assisted in recruitment operations and mentor evaluation within the ACM student chapter.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, SQL, Dart

AI & Machine Learning: PyTorch, TensorFlow, NumPy, Pandas, OpenCV, CNNs, Transformers, Computer Vision, Deep Learning, NLP, LLMs, LLM Fine-Tuning, QLoRA, RAG, LangChain, LangGraph, Multi-Agent Orchestration, Model Evaluation

Technologies/Frameworks: Flutter, NestJS, Fast API, SQLAlchemy, MongoDB, PostgreSQL, SQLite, REST APIs, Docker, Git/GitHub

Developer Tools: Jupyter, Google Colab, VS Code, IntelliJ, SQL Server Management Studio

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- IBM RAG and Agentic AI Specialization – Coursera & IBM Skills | 2026
- Machine Learning Specialization - Coursera & DeepLearning.AI | 2025
- Generative AI for Data Scientists Specialization - Coursera | 2025
- The Forward Program - McKinsey & Company | 2026
- The Science of Well-Being - Coursera & Yale University | 2025